

NVIDIA GPUs Instructions Energy Consumption

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Motivation

- ❑ **GPGPUs** are now powerful co-processors perform complex **general** calculations
 - Fundamental components in any high-performance computing (**HPC**) system
- ❑ Over the last decade, **NVIDIA** introduced **seven** different GPU architectures. Each has their microarchitecture and hardware characteristics
- ❑ The percentage of **disclosed characteristics** beyond what **NVIDIA** documents is small

- The structure and performance of their memory system
- Functionality of branch and memory divergence
- Instructions latencies and **energy consumption**



Contributions

- ✓ Accurate, and portable measurement of *NVIDIA GPUs* instructions energy consumptions
- ✓ Effect of *CUDA NVCC Compiler* optimizations on instructions' energy consumption
- ✓ Introduce & compare **three** *Software Techniques* to compute GPU kernel energy consumptions
- ✓ Verify different software techniques against **custom designed** *Hardware Power Measurement*
- ✓ Exhaustive comparison of all instructions in the ISA for different *NVIDIA GPUs* generations

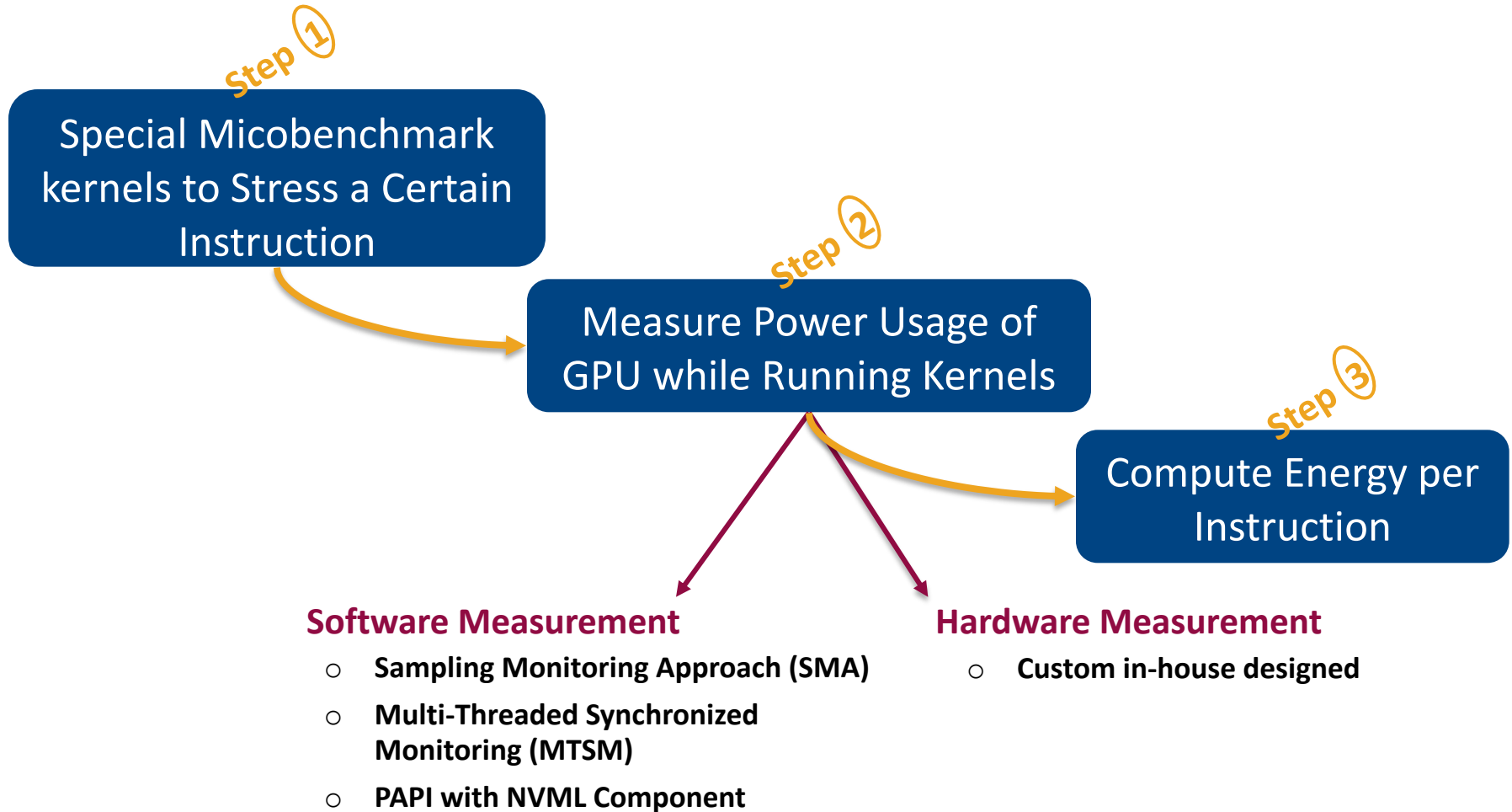
Why This Work Important!!

Gives concrete understanding of the underlying GPUs' microarchitecture

Makes energy measurements of any GPU kernel both efficient and accurate

Help in modeling the hardware (*ex: Arafa et. al*)

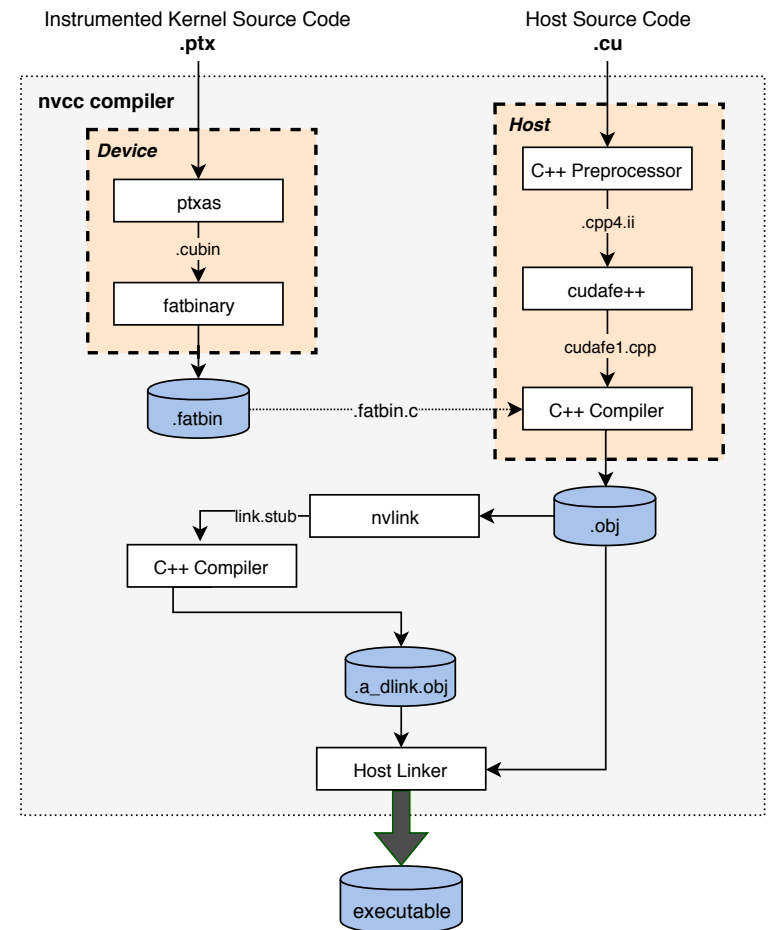
Methodology



PTX MicroBenchmarks

- ❑ *Parallel thread execution (PTX)* used to design the microbenchmarks kernels
- ❑ PTX allows us to control exact sequence of low-level instructions executing without any overhead
- ❑ Compatible with all NVIDIA architectures and generations

Compilation Workflow



PTX MicroBenchmarks

- ❑ The key idea is unrolling a loop and execute same instruction millions of times while recording the power then divide by number of instructions

CUDA Source Code (.cu)

```
__global__ void Div (int *c){}
void main(){
    ...
    ...
    dim3 gridSize = dim3(1);
    dim3 blockSize = dim3(1);

    Div<<gridSize, blockSize>>>(d_c);
    cudaDeviceSynchronize();
    ...
    ...
}
```

$$E_{instruction} = \frac{E_{total} - E_{overhead}}{\# instructions}$$

Instrumented PTX Kernel (.ptx)

```
.visible .entry Div(.param .u64 Div_param_0){

    .reg .b32    %r<15>;
    .reg .b64    %rd<5>;
    .reg .pred    %p<2>;
    ld.param.u64    %rd1, [Div_param_0];
    mov.u32         %r3, 3;
    mov.u32         %r4, 4;
    st.global.u32    [%rd4 + 12], 0;
    mov.u32         %r15, -1000000;

BB0_1:
    add.u32         %r4, %r4, 1;
    add.u32         %r3, %r3, 1;



```
div.u32 %r9, %r4, %r3;
div.u32 %r10, %r3, %r9;
div.u32 %r11, %r9, %r10;
div.u32 %r12, %r10, %r11;
div.u32 %r13, %r11, %r12;
```



```
ld.global.u32 %r9, [%rd4 + 12];
add.u32 %r10, %r9, %r13;
st.global.u32 [%rd4 + 12], %r10;
add.u32 %r15, %r15, 1;
setp.ne.u32 %p1, %r15, 0;
@%p1 bra BB0_1;
st.global.u32 [%rd4], %r15;
ret;
```


```

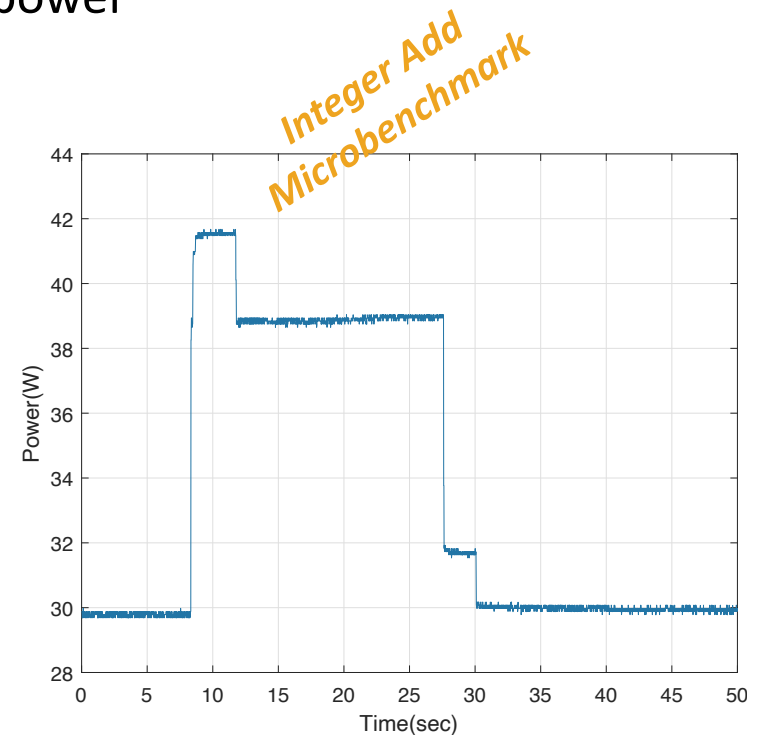
Remove to
compute the
overhead's
energy

Software Power Measurement

- ❑ Query onboard power sensors using NVIDIA Management Library
NVML API
- ❑ Different approaches can be used to query power

1) *Sampling Monitoring Approach (SMA)*

- Keep reading hardware sensor with a certain frequency
- Identifying exact window of kernel is hard and would affect output as power consumption varies through time

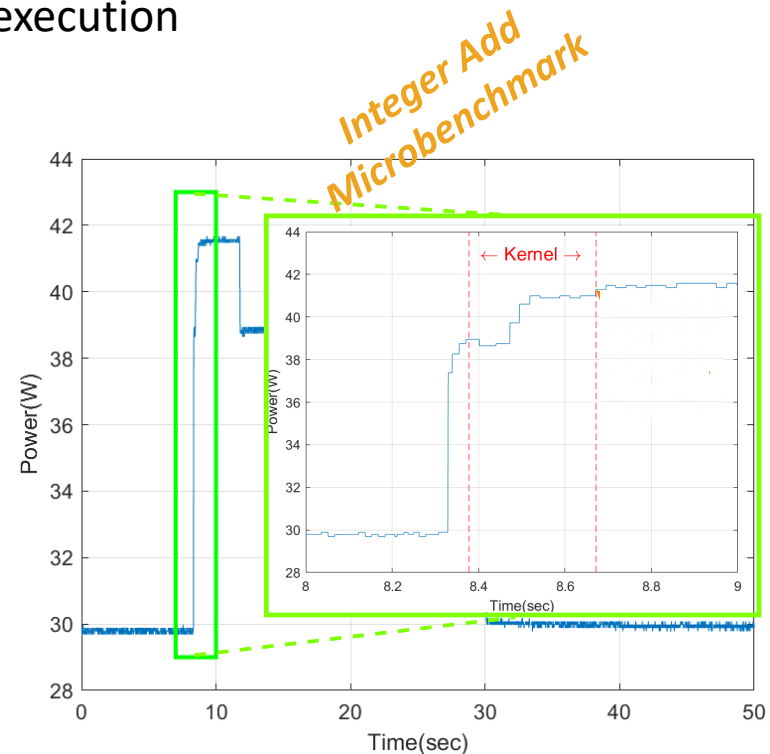


Software Power Measurement

2) Multi-Threaded Synchronized Monitoring (MTSM)

- Modify SMA to identify exact window of kernel execution
- **Pthreads** used for synchronization where one thread calls and monitors kernel while other thread records the power
- Area under curve is computed to get total energy consumption

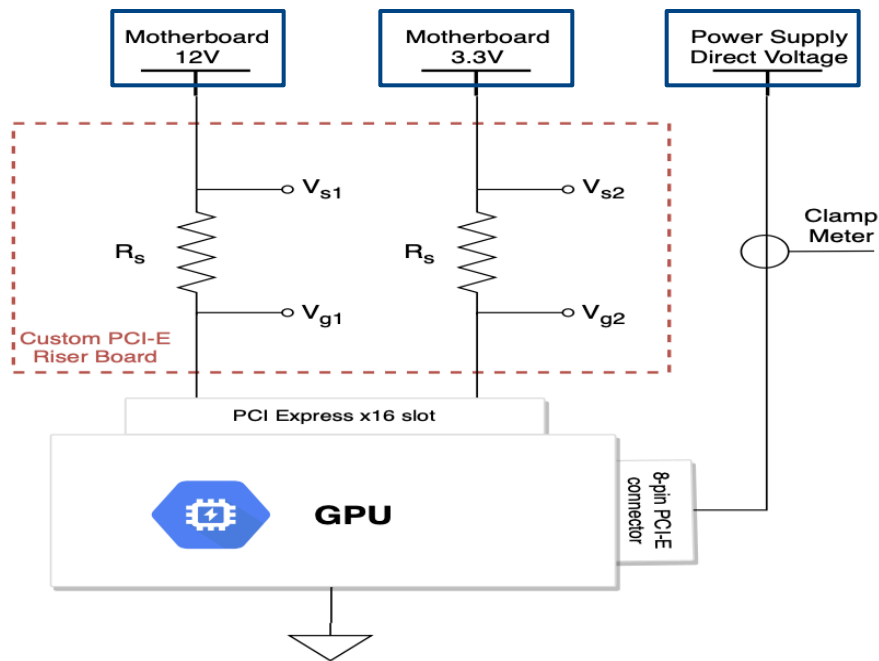
$$E = \frac{\text{Elapsed Time}}{\# \text{ Power Readings}} \times \sum_{i=0}^N \text{Power}_i$$



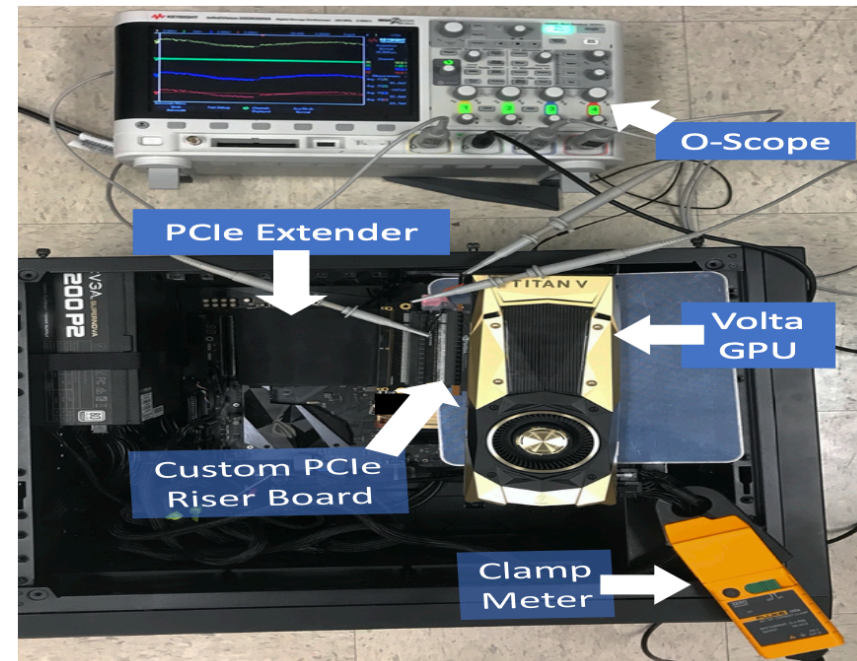
Hardware Power Measurement

- Designed a system to measure each GPU power sources at real time

Circuit Diagram



Hardware Setup w/ all Components



The hardware measurement is considered as the ground truth to verify the different software measurement techniques

Results

ALU Instructions Energy Consumption

- TITAN V (Volta) has best energy efficiency compared to other GPUs generations for different categories of the instructions.
- Overall, the energy of Non-Optimized is always more than the Optimized
- PAPI vs. MTSM: The dominant tendency of the results is that PAPI results are always more than the MTSM

Instruction	Optimized (μ J)				Non-Optimized (μ J)			
	TITAN X	1080 TI	TITAN V	TITAN RTX	TITAN X	1080 TI	TITAN V	TITAN RTX
Integer Arithmetic Instructions								
add/sub	0.046	0.020	0.0012	0.023	1.026	0.062	0.238	0.790
mul/mad	2.731	0.173	0.0014	0.028	4.296	3.599	0.4912	0.967
{s} div	6.782	4.989	4.066	6.549	64.631	44.561	27.058	48.494
{s} rem	6.683	4.760	4.014	6.798	61.300	42.052	25.117	47.907
Floating Single Precision Instructions								
add/sub	0.046	0.020	0.0012	0.028	1.026	0.6203	0.238	0.791
mul/mad	2.731	0.201	0.0014	0.281	4.296	0.6222	0.479	0.918
div	9.435	6.871	5.035	7.932	56.653	49.535	33.382	58.677
Double Precision Instructions								
add/sub	2.006	1.861	0.357	2.617	2.514	2.059	0.511	4.162
div	28.816	19.284	3.683	23.602	100.781	50.080	30.412	67.402
Special Mathematical Instructions								
rcl	5.342	4.032	2.427	3.951	18.283	13.146	10.054	19.260
sqr	5.39	4.007	2.422	4.053	18.669	13.412	10.270	19.698
sin/cos	0.151	0.274	0.586	0.906	0.894	0.865	0.978	1.883
Integer Intrinsic Instructions								
mul24()	0.294	0.273	0.212	0.341	6.426	4.864	2.325	4.594
sad()	0.015	0.1243	0.004	0.042	1.027	0.665	0.509	0.975

Conclusion

- ❑ In this paper, we accurately measure energy consumption of various PTX instructions that execute on NVIDIA GPUs.
 - ❑ We provide an in-depth comparison of various software techniques that query onboard internal GPU sensor
 - ❑ Overall, paper provides an easy and straightforward way *Multi-Threaded Synchronized Monitoring (MTSM)* that can be used to measure the energy consumption of any NVIDIA GPU kernel
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- ✓ If you have any questions, please contact me at:
yarafa@nmsu.edu
 - ✓ Source code is available at:
<https://github.com/NMSU-PEARL/GPUs-Energy>

more details!

